

# Yunhak Oh

[yunhak.oh@kaist.ac.kr](mailto:yunhak.oh@kaist.ac.kr) • [LinkedIn](#) • [Google Scholar](#) • [Github](#)

## PROFESSIONAL SUMMARY

Ph.D. candidate at KAIST developing machine learning methods to understand biology and disease. My research spans graph learning, spatial transcriptomics, and cellular modeling, with growing interests in disease modeling and patient heterogeneity. My long-term goal is to support therapeutic discovery and design, particularly RNA therapeutics.

I am interested in research or applied scientist roles and future postdoctoral opportunities in AI for biology, where I can bring my research and industry experience to biomedical discovery.

## TECHNICAL AREAS

### Artificial Intelligence for Science

- Cellular Representation Learning & Transcriptomics
- Disease Modeling & Patient Heterogeneity
- Therapeutic Discovery & RNA Therapeutics

## PROFESSIONAL EXPERIENCE

### HITS, Seoul, South Korea

- Research Intern, AI Research 1 Team Nov 2025 – May 2026
  - Engineering a novel single-cell foundation model by leveraging large-scale scRNA-seq datasets (e.g., 80M+ cells) to simulate cellular dynamics within the 'Virtual Cell' project.

### NielsenIQ (formerly Nielsen), Seoul, South Korea

- Manager, Data Science Jul 2018 – Aug 2021
  - Reduced operational costs by \$54K USD in three months by spearheading an auto-coding project that used ML models to classify web-crawled product descriptions.
  - Generated \$71.9K USD in new revenue by engineering a novel e-commerce analysis model that integrated disparate retailer data to more accurately capture market growth.
  - Enabled a \$901K USD M&A project by spearheading the critical data and solutions integration between the two merging companies.
  - As Technical Lead, directed a global initiative to automate client inquiry resolution, driving major operational efficiency gains across international teams.
- Senior Executive, Data Science Jul 2017 – Jun 2018
  - Cut production time by 83% by developing and implementing a data-driven methodology for historical data estimation.
  - Modernized trade practices by orchestrating the transition to modern retail point-of-sale (POS) systems, enhancing data accuracy and operational readiness.
- Executive, Data Science Jan 2015 – Jun 2017
  - Slashed reporting and data extraction times by 50% and 92% respectively by proactively developing a new suite of software automation tools.

## EDUCATION

### KAIST (Korea Advanced Institute of Technology), Daejeon, South Korea

- Ph.D. in Graduate School of Data Science Sep 2023 – Present
  - Research Interest: Recommender System, Graph Representation Learning, AI4Science (Cell Biology)
  - Adviser: [Prof. Chanyoung Park](#)
- M.S. in Industrial & Systems Engineering Sep 2021 – Aug 2023
  - Research Interest: Recommender System, Graph Representation Learning
  - Adviser: [Prof. Chanyoung Park](#)

### SungKyunKwan University, Gyeonggi, South Korea

Mar 2009 – Feb 2015

- B.S.E. in System Management Engineering
  - Graduated 1st in the department among 133 students
  - Included two years of mandatory military service in the Office of the President of the Republic of Korea
- B.A. in Psychology
  - Dual Degree

## PUBLICATIONS

(\*: Equal contribution)

## CONFERENCES

- [C9] scTrilemma: Balancing Identity, Invariance, and Fidelity in Single-Cell Representation Learning  
**Yunhak Oh**, Yoonho Lee, Junseok Lee, Namkyeong Lee, Sang-Yeon Hwang, Yinhua Piao, Hyomin Kim, Seonghwan Kim, Jaechang Lim, Woo Youn Kim, Sungsoo Ahn, Chanyoung Park  
**NeurIPS 2026** - The Fortieth Annual Conference on Neural Information Processing Systems  
**and NeurIPS 2026 Workshop** - The Third Workshop on GenAI for Health

- [C8] Screening Lipid Nanoparticles through Structure-Ratio Alignment  
Yoonho Lee, **Yunhak Oh**, Hoyoung Choi, Chanyoung Park  
**NeurIPS 2026** - The Fortieth Annual Conference on Neural Information Processing Systems
- [C7] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction  
Yinhua Piao, Hyomin Kim, Seonghwan Kim, **Yunhak Oh**, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, Chanyoung Park, Sungsoo Ahn  
**ICML 2026** - The International Conference on Machine Learning  
*and ICLR 2026 Workshop* - Machine Learning for Genomics Explorations  
*and ICLR 2026 Workshop* - Generative and Experimental Perspectives for Biomolecular Design
- [C6] 3D Interaction Geometric Pre-training for Molecular Relational Learning  
Namkyeong Lee, **Yunhak Oh**, Heewoong Noh, Gyoung S. Na, Tianfan Fu, Chanyoung Park  
**NeurIPS 2025 (Spotlight)** - The Thirty-Ninth Conference on Neural Information Processing Systems  
*and NeurIPS 2024 Workshop* - AI for New Drug Modalities
- [C5] Oldie but Goodie: Re-illuminating Label Propagation on Graphs with Partially Observed Features  
Sukwon Yun, Xin Liu, **Yunhak Oh**, Junseok Lee, Tianlong Chen, Tsuyoshi Murata, Chanyoung Park  
**KDD 2025** - ACM SIGKDD Conference on Knowledge Discovery and Data Mining
- [C4] Global Context-aware Representation Learning for Spatially Resolved Transcriptomics  
**Yunhak Oh\***, Junseok Lee\*, Yeongmin Kim, Sangwoo Seo, Namkyeong Lee, Chanyoung Park  
**ICML 2025** - The International Conference on Machine Learning
- [C3] Subgraph Federated Learning for Local Generalization  
Sungwon Kim, Yoonho Lee, **Yunhak Oh**, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, Chanyoung Park  
**ICLR 2025 (Oral, top 1.8%)** - The International Conference on Learning Representations  
*and KDD 2024 Workshop (Oral, Best Paper Award)* - Federated Learning for Data Mining and Graph Analytics (FedKDD)
- [C2] MUSE: Music Recommender System with Shuffle Play Recommendation Enhancement  
**Yunhak Oh\***, Sukwon Yun\*, Dongmin Hyun, Sein Kim, Chanyoung Park  
**CIKM 2023** - ACM International Conference on Information and Knowledge Management
- [C1] GraFN: Semi-Supervised Node Classification on Graph with Few Labels via Non-Parametric Distribution Assignment  
Junseok Lee, **Yunhak Oh**, Yeonjun In, Namkyeong Lee, Dongmin Hyun, Chanyoung Park  
**SIGIR 2022** - ACM SIGIR Conference on Research and Development in Information Retrieval (Short paper)

#### JOURNALS

- [J2] Discovering relationships between skin type and life style using data mining techniques: A case study of Korea  
Taeheung Kim, Jihyun Ha, Jong-Seok Lee, **Yunhak Oh**, Yong Ju Cho  
Industrial Engineering and Management Systems (2016.03)
- [J1] Using data mining techniques to predict win-loss in Korean professional baseball games  
**Yunhak Oh**, Han Kim, Jaesub Yun, Jong-Seok Lee  
Journal of Korean Institute of Industrial Engineers (2014.02)

#### WORKSHOPS

- [W1] Progressive Multi-Agent Reasoning for Biological Perturbation Prediction  
Hyomin Kim, Sang-Yeon Hwang, Jaechang Lim, Yinhua Piao, **Yunhak Oh**, Woo Youn Kim, Chanyoung Park, Sungsoo Ahn, Junhyeok Jeon  
**ICML 2026 Workshop** - Generative and Agentic AI for Biology

#### AWARDS & SCHOLARSHIPS

- Best Paper Award** 2024  
▪ KDD 2024 Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD), Barcelona, Spain
- Nielsen Simply Excellent Awards, NielsenIQ**  
▪ **Gold Award**, Developed and rolled out a Client Inquiry Tool for the global market 2020  
▪ **Gold Award**, Created a best practice of Digitalization and Automation 2020

- **Silver Award**, Developed a Client Inquiry Automation tool 2019
- **Platinum Award**, Developed and rolled out auto-coding project 2019
- **Gold Award**, Contributed data and solution integration in the M&A process 2018
- **Gold Award**, Launched E-commerce Market Read Index version 3.0 of South Korea 2018
- **Gold Award**, Led Digitalization and Automation project 2017
- **Gold Award**, Enhanced Ice-cream Market Read Index of South Korea 2017
- **Gold Award**, Enhanced FMCG Market Read Index of South Korea and boosted client satisfaction 2015

**Certificate, Nielsen** 2019

- Selected as one of the top 20 global data science talents to participate in a leadership development program

**Departmental Excellence Award, SungKyunKwan University** 2015

- Presented to the top-ranked graduate as the department's representative at the commencement ceremony

**National Science and Engineering Scholarship, Korea Student Aid Foundation** 2013 – 2014

- Awarded to a top student in the Department of System Management Engineering

**Bronze Award, Korea Institute of Industrial Engineers** 2013

- 3rd place, Solved industrial problems by building an ML model at a University Student Project Competition

**Academic Excellence Scholarship, SungKyunKwan University** 2009 – 2011

## PROFESSIONAL SERVICES

### ORGANIZING COMMITTEE

- MM4SciReal: Multimodal AI for Science and the Real World (Workshop @ ACCV) 2026

### PROGRAM COMMITTEE/REVIEWER

- Conference on Neural Information Processing Systems (NeurIPS) 2026
- International Conference on Learning Representations (ICLR) 2025 – Present
- AI4Science Workshop @ NeurIPS / ICML 2025 – Present
- Conference on Information and Knowledge Management (CIKM) - Short & Applied Research 2025

## TALKS AND SEMINARS

### MUSE: Music Recommender System with Shuffle Play Recommendation Enhancement

- Top Conference Session of Korea Software Congress (KSC) 2023

## REFERENCES

**Prof. Chanyoung Park**, Associate Professor, KAIST

Email: cy.park@kaist.ac.kr

**Prof. Jong-Seok Lee**, Associate Professor, KAIST

Email: jongseok.lee@kaist.ac.kr

[CV compiled on 2026-10-05]